MILITARY SPECIFICATION

INSULATION, PANEL, NOISE- AND THERMAL-BARRIER

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers insulation with sound- and thermal-barrier properties for use on metal or plastic panels.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810D	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-45662	- Calibration System Requirements.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C177	- Steady-State Thermal Transmission Properties by Means of the Guarded Hot Plate.
ASTM D297	- Rubber Products - Chemical Analysis.
ASTM D412	- Rubber Properties in Tension.
ASTM D624	- Rubber Property - Tear Resistance.
ASTM D903	- Peel or Stripping Strength of Adhesive Bonds.
ASTM D3574	- Flexible Cellular Materials - Slab, Bonded, and Molded Urethane Foams.
ASTM E90	- Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J369	- Flammability of Automotive Interior Material - Horizontal Test Method.

(Information as to the availability of the above standard may be obtained from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish insulation samples which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be a sample of the insulation to be furnished to the Government. All subsequent insulation delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. The materials shall be as specified herein. The insulation shall not contain any materials which would cause adverse effects on the health of the personnel (see 4.8.1).

3.2.1 Composition. The insulation shall be a composite consisting of barrier and foam. The barrier shall completely envelope the foam or, if not, the foam shall be closed cell.

3.2.1.1 Barrier. The barrier shall be 0.8 pound per square foot $\pm$ 10 percent loaded plastic having a density of 2.5 pounds per cubic foot. The barrier shall be self-healing puncture-resistant (see 4.8.1).

3.2.1.2 Foam. The foam shall be 0.50-inch-thick urethane foam (see 4.8.1).

3.2.2 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials; however, all materials shall be new and unused. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

3.3 Design and construction.

3.3.1 Configuration. The configuration of the insulation shall be as specified on the applicable drawing (see 4.8.2).

3.3.2 Color. The color of the insulation shall be as specified on the applicable drawing (see 4.8.2).

3.4 Performance.

3.4.1 Adhesion strength. The adhesion strength of the barrier to the foam at a 180 degree peel shall be not less than 2.2 pounds per linear inch (see 4.8.3).

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3.4.2 Sound transmission loss. The sound transmission loss of the insulation mounted on 16-gauge sheet steel shall be as specified in table I (see 4.8.4).

TABLE I. Acoustical performance.

Frequency (hertz)	Sound transmission loss (decibels)
250	29
500	34
1000	40
2000	45
4000	55

3.4.3 Flammability. The burning rate of the insulation shall be not greater than 3 inches per minute (see 4.8.5).

3.4.4 Environmental.

3.4.4.1 High temperature. The insulation shall remain physically stable when exposed to a temperature of 300 degrees Fahrenheit (°F) (see 4.8.6).

3.4.4.2 Low temperature. The insulation shall remain physically stable when exposed to a temperature of -40°F (see 4.8.7).

3.4.5 Barrier.

3.4.5.1 Tear resistance. The tear resistance of the barrier shall be not less than 20 pounds force per inch (ppi) (see 4.8.8).

3.4.5.2 Tensile strength. The tensile strength of the barrier shall be not less than 120 pounds force per square inch (psi) (see 4.8.9).

3.4.6 Foam.

3.4.6.1 Thermal conductivity. The thermal conductivity of the foam shall be not greater than 0.30 British thermal unit (Btu)/[(hour)(square foot)(°F/inch)] (see 4.8.10).

3.4.6.2 Density. The density of the foam shall be 2.00 ± 0.2 pounds per cubic foot (see 4.8.11).

3.4.6.3 Tensile strength. The tensile strength of the foam shall be not less than 15 psi (see 4.8.12).

3.4.6.4 Elongation. The elongation of the foam shall be not less than 200 percent (see 4.8.13).

3.4.6.5 Compression set. The compression set of the foam shall be not greater than 10 percent at 70 ± 5°F (see 4.8.14).

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3.5 Identification marking. Identification marking shall be in accordance with MIL-STD-130 (see 4.5.2 and 4.8.2).

3.6 Workmanship. Workmanship shall be of high quality and in accordance with high-grade practices. Insulation shall be uniform in quality and free of foreign matter, cracks, tears, cuts, or any other defect that will affect its life, serviceability, or appearance. Workmanship shall be such as to ensure insulation which is in conformance to this specification (see 4.5.2 and 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply wih all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

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a. Air temperature 73 ± 18°F
b. Barometric pressure 28.5 + 2.0 inches mercury (Hg)
- 3.0
c. Relative humidity 50 ± 30 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select insulation samples produced under the production contract (see 6.2) for first article inspection. First article samples shall be inspected as specified in table II. Approval of first article samples by the Government shall not relieve the contractor of his obligation to supply insulation that is fully representative of those inspected as first article samples. Any changes or deviation of the production units from the first article samples shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, first article inspection shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that the insulation already produced prior to completion of first article inspection is similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Materials	3.2 thru 3.2.1.2	4.8.1	X		X	
Defects (see 4.5.2 and table III)	3.3.1, 3.3.2, 3.5 and 3.6	4.8.2	X	X		
Adhesion strength	3.4.1	4.8.3	X			X
Sound transmission loss	3.4.2	4.8.4	X			X
Flammability	3.4.3	4.8.5	X			X
High temperature	3.4.4.1	4.8.6	X			X
Low temperature	3.4.4.2	4.8.7	X			X
Tear resistance	3.4.5.1	4.8.8	X			X
Tensile strength	3.4.5.2	4.8.9	X			X
Thermal conductivity	3.4.6.1	4.8.10	X			X
Density	3.4.6.2	4.8.11	X			X
Tensile strength	3.4.6.3	4.8.12	X			X
Elongation	3.4.6.4	4.8.13	X			X
Compression set	3.4.6.5	4.8.14	X			X

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4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the material manufactured an identifiable production period (not exceeding 24 hours), from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 8 consecutively produced insulation samples that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs.

4.5.1.3 Sampling for tests. Samples for acceptance tests shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimension, out of tolerance (see 3.3.1).	SIE 1/
102	Identification marking, improper (see 3.5).	Visual
103	Faulty workmanship affecting performance (see 3.6).	Visual
Minor	AQL 2.5% Defective	
201	Color not as specified (see 3.3.2).	Visual
202	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

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4.5.3 Tests. Each insulation sample shall be subjected to the quality conformance tests specified in table II.

4.6 Control tests. Control tests shall be conducted on 1 insulation sample from each lot of 200 units consecutively produced, except that not more than 1 test shall be performed in a 3-month period, nor less than 1 test in a 6-month period. The insulation shall be subjected to the tests specified in table II.

4.7 Failure. Failure of any sample to pass any of the specified quality conformance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials. Conformance to 3.2 through 3.2.1.2 shall be determined by inspection of contractor records providing proof or certification that processing and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3.1, 3.3.2, 3.5, and 3.6 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with SIE.

4.8.3 Adhesion strength. To determine conformance to 3.4.1, the adhesion strength of the barrier to the foam shall be tested as specified in ASTM D903.

4.8.4 Sound transmission loss. To determine conformance to 3.4.2, the insulation shall be tested as specified in ASTM E90.

4.8.5 Flammability. To determine conformance to 3.4.3, the insulation shall be tested as specified in SAE J369.

4.8.6 High temperature. To determine conformance to 3.4.4.1, the insulation shall be tested as specified in MIL-STD-810D, method 501.2, procedure I, at 300°F for seven 24-hour periods.

4.8.7 Low temperature. To determine conformance to 3.4.4.2, the insulation shall be tested as specified in MIL-STD-810D, method 502.2, procedure I, at -40°F for 4 hours.

4.8.8 Tear resistance. To determine conformance to 3.4.5.1, the barrier shall be tested as specified in ASTM D624, die C.

4.8.9 Tensile strength. To determine conformance to 3.4.5.2, the barrier shall be tested as specified in ASTM D412, die A.

4.8.10 Thermal conductivity. To determine conformance to 3.4.6.1, the foam shall be tested as specified in ASTM C177.

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4.8.11 Density. To determine conformance to 3.4.6.2, the foam shall be tested as specified in ASTM D297.

4.8.12 Tensile strength. To determine conformance to 3.4.6.3, the foam shall be tested as specified in ASTM D3574, die A.

4.8.13 Elongation. To determine conformance to 3.4.6.4, the foam shall be tested as specified in ASTM D412.

4.8.14 Compression set. To determine conformance to 3.4.6.5, the foam shall be tested as specified in ASTM D3574.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. This insulation is intended to be used as panel insulation in military vehicles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article samples are not required (see 3.1).
c. Title, number, and date of applicable drawing (see 3.3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article inspection is not required (see 4.4).
h. Number of samples to be selected for first article inspection (see 4.4).
i. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or

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industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Insulation
Noise barrier
Panel insulation
Thermal barrier

6.5 Supersession data. This military specification supersedes AM General Corporation specification AMG 193 revision A, dated 16 May 1984.

Custodian:
Army - AT

Preparing activity:
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(Project No. 5640-A445)

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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